

# **WINNER ENGINEERING LEARNING HUB SERIES**

INTERMEDIATE TOPIC

DUCT, PIPE &

EQUIPMENT

SUPPORTING DETAILS



Winner Engineering Pte Ltd  
永胜工程私人有限公司  
Excellence in Service

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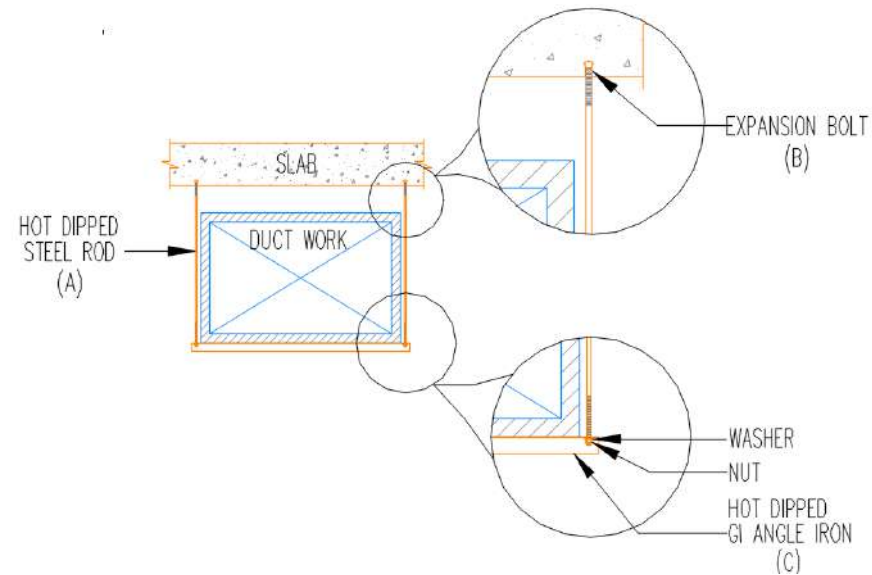
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# Duct Supporting Detail:

## GI DUCT:

1. Rectangular Duct Support
2. Round Duct Support

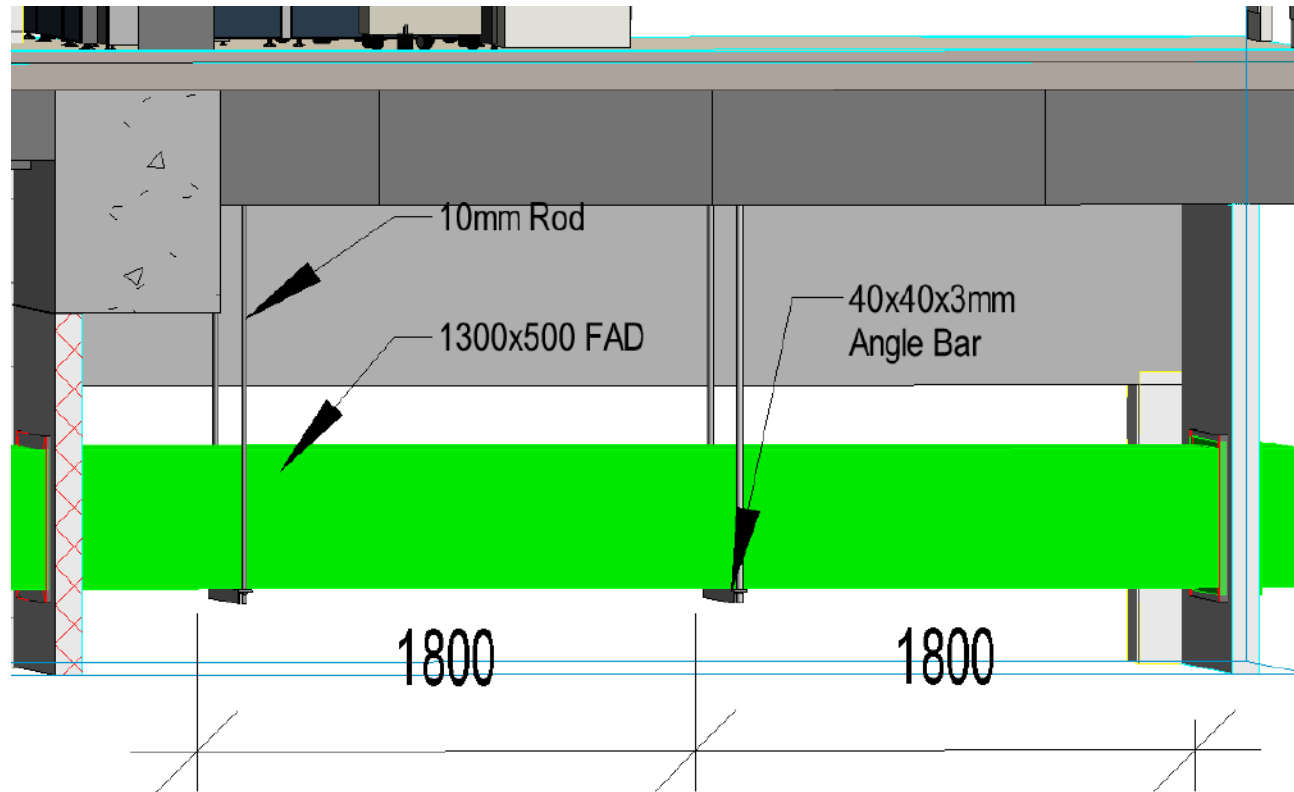
### 1.Rectangular Duct Support:



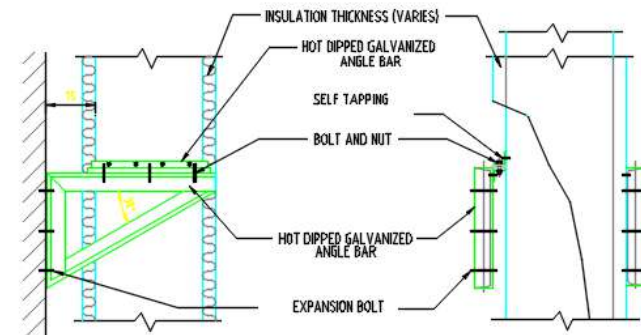
## SCHEDULE FOR SUPPORT ROD & ANGLES

| DIMENSION OF LONGEST SIDE OF DUCT(mm) | ROD SIZE (A) | EXPANSION BOLT (B)       | GI SUPPORT (C)    | INTERVAL OF SUPPORT |
|---------------------------------------|--------------|--------------------------|-------------------|---------------------|
| UP TO 300                             | DIA. 6mm     | DIA. 6mm EXPANSION BOLT  | 20x20x3 ANGLE BAR | 2.4m                |
| 325 TO 750                            | DIA. 6mm     | DIA. 6mm EXPANSION BOLT  | 25x25x3 ANGLE BAR | 1.8m                |
| 775 TO 1500                           | DIA. 10mm    | DIA. 10mm EXPANSION BOLT | 40x40x3 ANGLE BAR | 1.8m                |
| 1525 TO 2100                          | DIA. 10mm    | DIA. 10mm EXPANSION BOLT | 50x50x5 ANGLE BAR | 1.8m                |
| 2125 OVER                             | DIA. 10mm    | DIA. 10mm EXPANSION BOLT | 50x50x6 ANGLE BAR | 1.5m                |

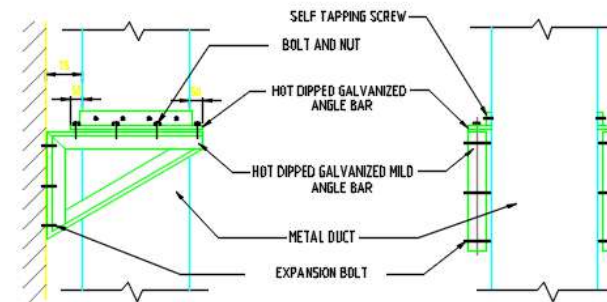
# 1. Rectangular Duct:



## Riser Duct:

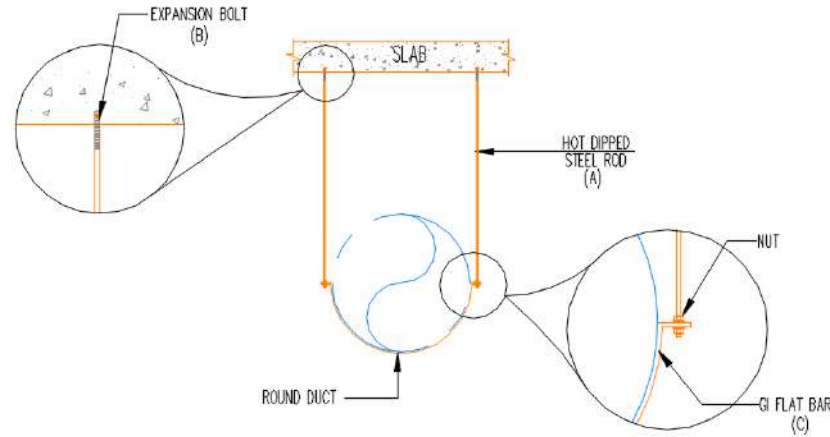


WALL BRACKET FOR INSULATED DUCT



WALL BRACKET FOR NON-INSULATED DUCT

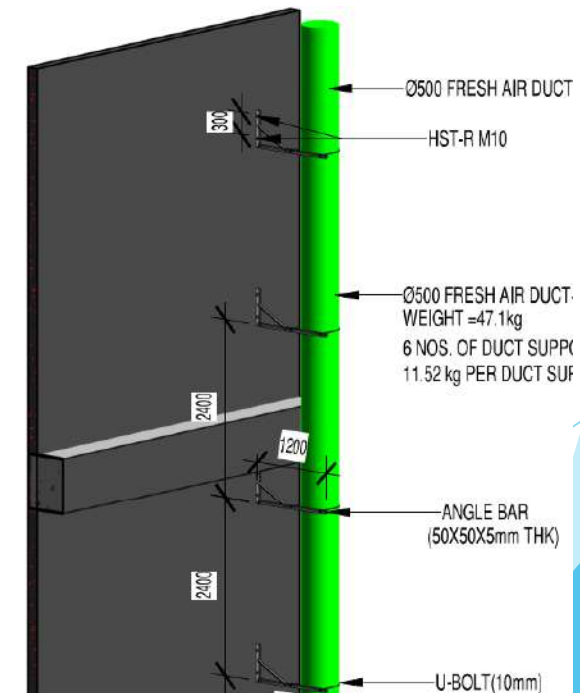
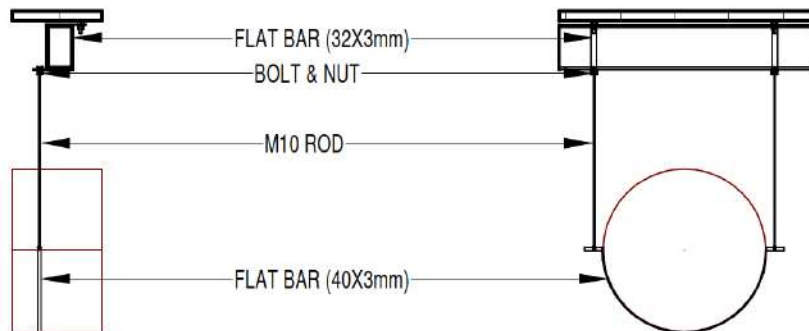
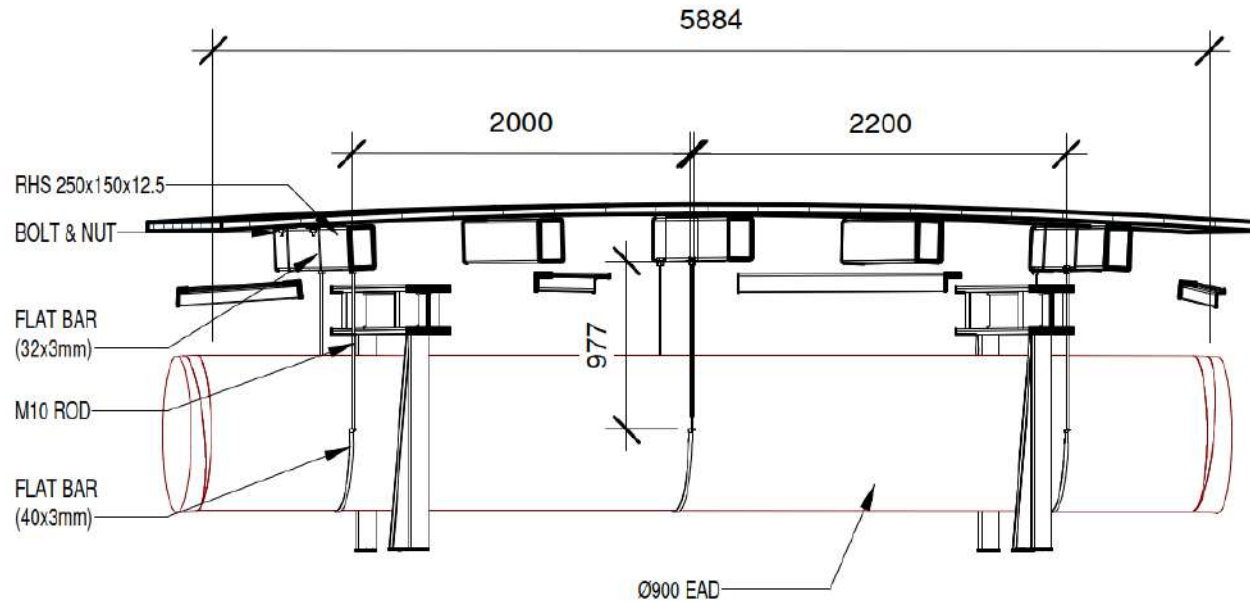
## 2.Round Duct



### SCHEDULE FOR SUPPORT ROD & ANGLES

| DIAMETER OF DUCT(mm) | ROD SIZE<br>(A) | EXPANSION BOLT<br>(B)    | FLAT BAR<br>(C) | INTERVAL OF<br>SUPPORT |
|----------------------|-----------------|--------------------------|-----------------|------------------------|
| UP TO 250            | DIA. 6mm        | DIA. 6mm EXPANSION BOLT  | 40X3 mm         | 2.4m                   |
| 450                  | DIA. 6mm        | DIA. 6mm EXPANSION BOLT  | 40X3 mm         | 2.4m                   |
| 600                  | DIA. 6mm        | DIA. 6mm EXPANSION BOLT  | 40X3 mm         | 2.4m                   |
| 900                  | DIA. 10mm       | DIA. 10mm EXPANSION BOLT | 40X3 mm         | 2.4m                   |

# Round Duct:



PID DUCT:

## SCHEDULE FOR SUPPORT ROD & ANGLES

| DIMENSION OF LONGEST SIDE OF PID DUCT(mm) | ROD SIZE (A) | EXPANSION BOLT (B)       | GI SUPPORT (C)    | INTERVAL OF SUPPORT |
|-------------------------------------------|--------------|--------------------------|-------------------|---------------------|
| UP TO 300                                 | DIA. 6mm     | DIA. 6mm EXPANSION BOLT  | 20x20x3 ANGLE BAR | 2.4m                |
| 325 TO 750                                | DIA. 6mm     | DIA. 6mm EXPANSION BOLT  | 25x25x3 ANGLE BAR | 1.8m                |
| 775 TO 1500                               | DIA. 10mm    | DIA. 10mm EXPANSION BOLT | 40x40x3 ANGLE BAR | 1.8m                |
| 1525 TO 2100                              | DIA. 10mm    | DIA. 10mm EXPANSION BOLT | 50x50x5 ANGLE BAR | 1.8m                |
| 2125 OVER                                 | DIA. 10mm    | DIA. 10mm EXPANSION BOLT | 50x50x6 ANGLE BAR | 1.5m                |



# Duct Weight Calculation :

Total duct weight= Duct weight + Insulation weight

Duct weight= Volume x Density

PID Density = 60 kg/m<sup>3</sup>

Insulation weight= Insulation Volume x Insulation Density

Insulation Density= 32 kg/m<sup>3</sup>

## SCHEDULE FOR DUCT WEIGHT

| GI SHEET THICKNESS (MM) | BASIC MASS (KG/MM <sup>2</sup> ) | UNIT MASS BASE METAL (KG/M <sup>2</sup> ) | COATING MASS CONSTANT (KG/MM <sup>2</sup> ) | UNIT MASS OF CUT-LENGTH (KG/M <sup>2</sup> ) |
|-------------------------|----------------------------------|-------------------------------------------|---------------------------------------------|----------------------------------------------|
| 0.6                     | 7.85                             | 4.71                                      | 0.282                                       | 4.992                                        |
| 0.8                     |                                  | 6.28                                      |                                             | 6.562                                        |
| 1.0                     |                                  | 7.85                                      |                                             | 8.132                                        |
| 1.2                     |                                  | 9.42                                      |                                             | 9.702                                        |

| GI DUCT SIZE (mm) | UNIT MASS (KG/M <sup>2</sup> ) | DUCT WEIGHT (KG/M) | INSULATION WEIGHT (KG/M) | TOTAL WEIGHT (KG/M) |
|-------------------|--------------------------------|--------------------|--------------------------|---------------------|
| 300 x 300         | 4.992                          | 5.99               | 2.88                     | 8.87                |
| 750 x 750         | 6.562                          | 19.69              | 7.20                     | 26.89               |
| 1500 x 1500       | 8.132                          | 48.79              | 14.40                    | 63.19               |
| 2000 x 2000       | 9.702                          | 77.62              | 19.20                    | 96.82               |

| SS DUCT SIZE (mm) | UNIT MASS (kg/M <sup>2</sup> ) | DUCT WEIGHT (KG/M) | INSULATION WEIGHT (KG/M) | TOTAL WEIGHT (KG/M) |
|-------------------|--------------------------------|--------------------|--------------------------|---------------------|
| 300 x 300         | 7.666                          | 9.19               | 2.88                     | 12.07               |
| 750 x 750         |                                | 22.99              | 7.20                     | 30.19               |
| 1500 x 1500       | 9.432                          | 56.59              | 14.40                    | 63.19               |
| 2000 x 2000       |                                | 75.45              | 19.20                    | 94.65               |

| ROUND DUCT GAUGE TABLE<br>(Spiral Seam / Unreinforced) |                                                    |
|--------------------------------------------------------|----------------------------------------------------|
| GI Sheet Thickness                                     | Max. Duct Size (Dia. mm)<br>500 Pa Static Pressure |
| 0.5 mm                                                 | UP TO 380                                          |
| 0.6 mm                                                 | 381 – 450                                          |
| 0.8 mm                                                 | 451 – 650                                          |
| 1.0 mm                                                 | 651 – 1220                                         |
| 1.2 mm                                                 | 1221 – 1520                                        |

| 1000 Pa STATIC POS. or NEG. RECTANGULAR DUCT CONSTRUCTION<br>With Tie Rod Reinforcement<br>1200-mm TDF (Finished length 1100mm)<br>JTR-Joint Tie Rod CTR-Center Tie Rod |              |         |       |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------|-------|------------|
| GI SHEET THICKNESS<br>DUCT SIZE                                                                                                                                         | 1.2mm        | 1.0mm   | 0.8mm | 0.6mm      |
| UP TO 350                                                                                                                                                               |              |         |       | SLIP JOINT |
| 351 TO 450                                                                                                                                                              |              |         |       | TDF        |
| 451 TO 750                                                                                                                                                              |              |         | TDF   |            |
| 751 TO 900                                                                                                                                                              |              | TDF     |       |            |
| 901 TO 1050                                                                                                                                                             |              | TDF/CTR |       |            |
| 1051 TO 1350                                                                                                                                                            | TDF/CTR      |         |       |            |
| 1351 TO 1800                                                                                                                                                            | TDF/CTR/JTR  |         |       |            |
| 1801 & ABOVE                                                                                                                                                            | NOT DESIGNED |         |       |            |

# CHILLED WATER PIPE :

TABLE-1

| A \ B | 15-80                       | 100                         | 125                        | 150                         | 200                         | 250                         | 300                         |
|-------|-----------------------------|-----------------------------|----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
| 15-80 | D-10<br>L-50X50X5<br>(1 PC) | D-13<br>L-75X40X5<br>(1 PC) | D-13<br>L-75X40X5<br>(1PC) | D-13<br>L-75X40X5<br>(1PC)  | D-16<br>L-75X40X5<br>(2PCS) | D-16<br>L-75X40X5<br>(2PCS) | D-16<br>L-75X40X5<br>(2PCS) |
| 100   |                             | ↓                           |                            | D-16<br>L-75X40X5<br>(2PCS) |                             |                             |                             |
| 125   |                             |                             | ↓                          |                             |                             |                             |                             |
| 150   |                             |                             |                            | ↓                           |                             |                             |                             |
| 200   |                             |                             |                            |                             | ↓                           |                             |                             |
| 250   |                             |                             |                            |                             |                             | ↓                           |                             |
| 300   |                             |                             |                            |                             |                             |                             | ↓                           |

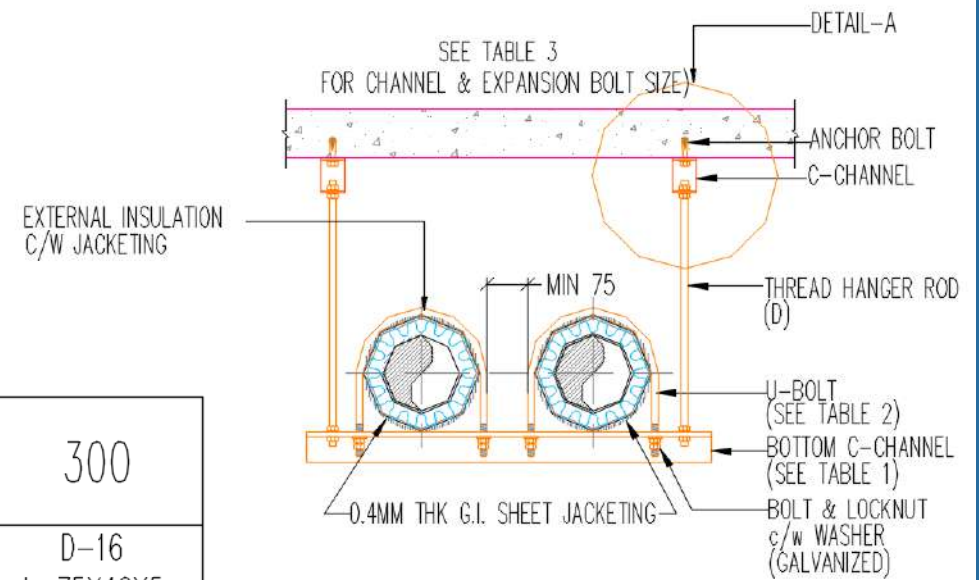
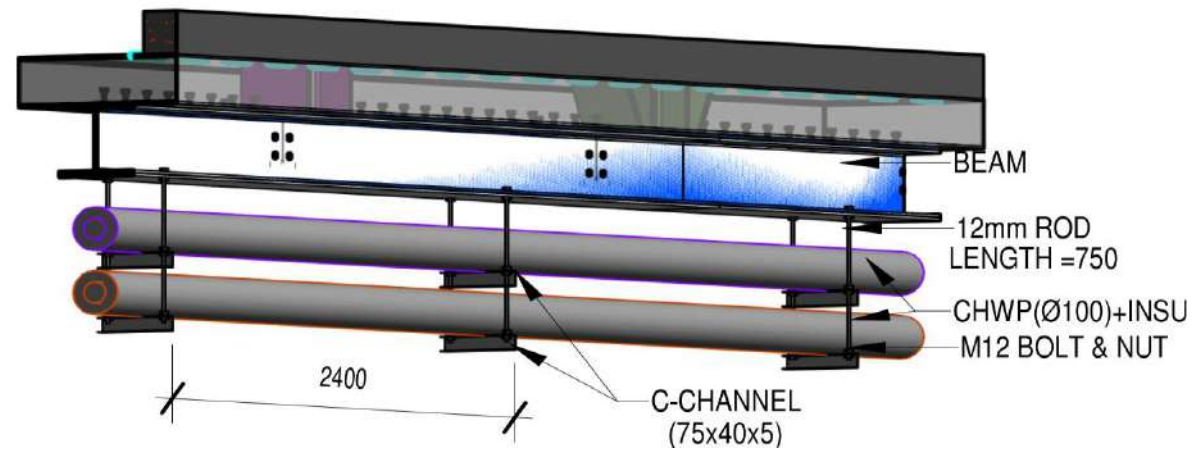


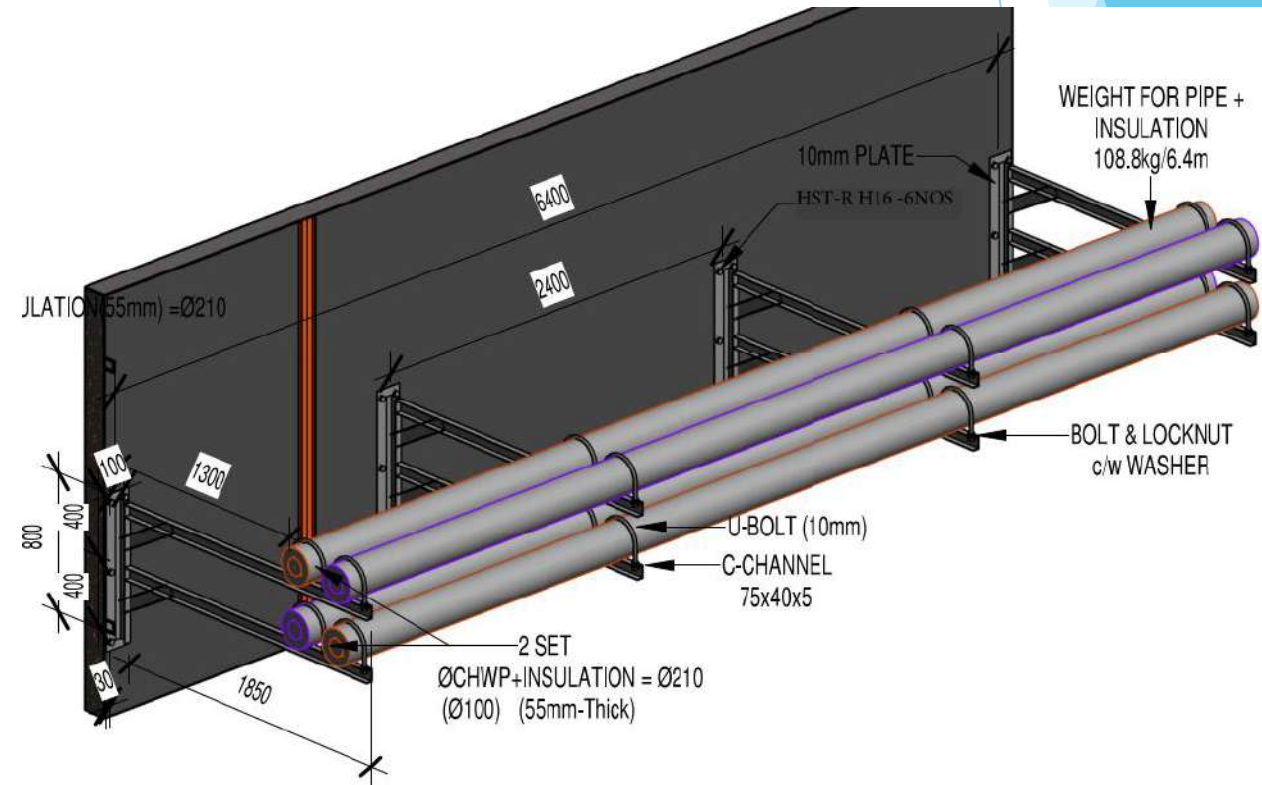
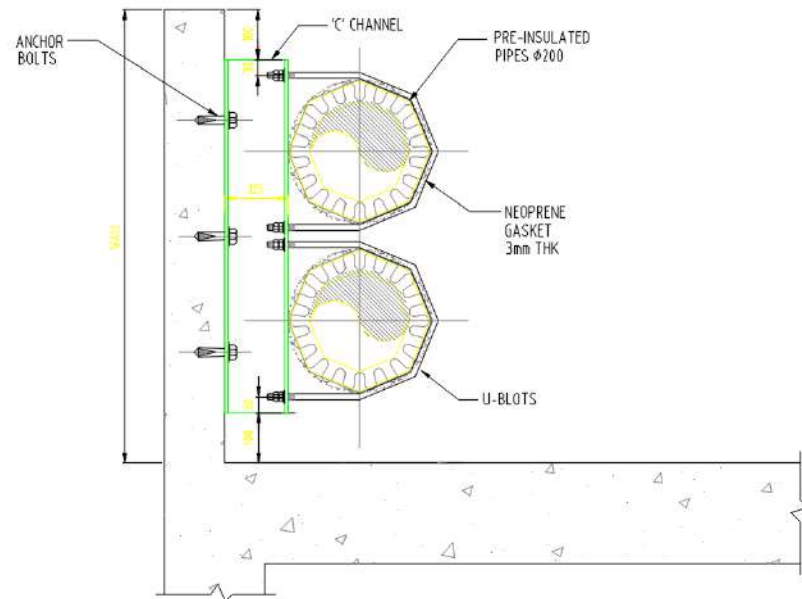
TABLE-2

| PIPE SIZE (NORMAL) (MM) | SIZE OF U-BOLT (MM) |
|-------------------------|---------------------|
| 15 TO 80                | 8                   |
| 100 TO 125              | 10                  |
| 150 TO 300              | 12                  |

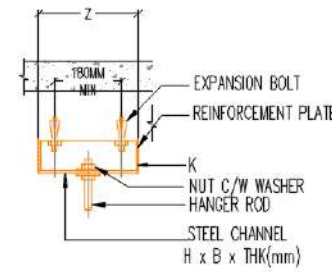
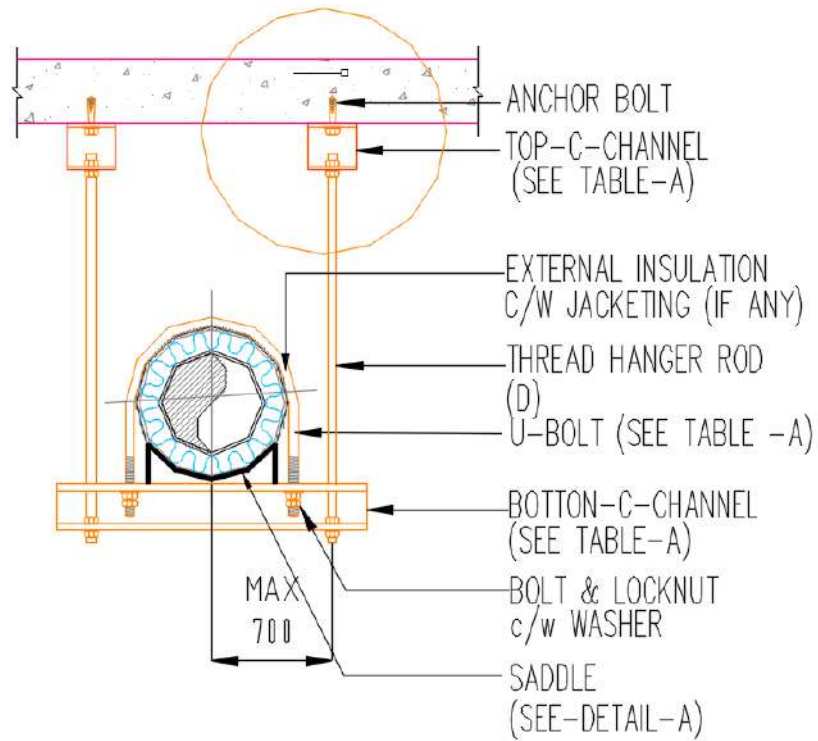
# CHILLED WATER PIPE :



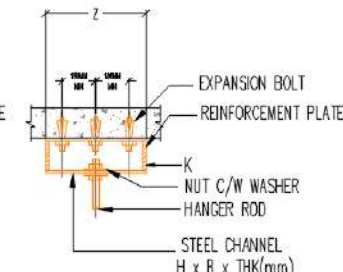
## RISER PIPE:



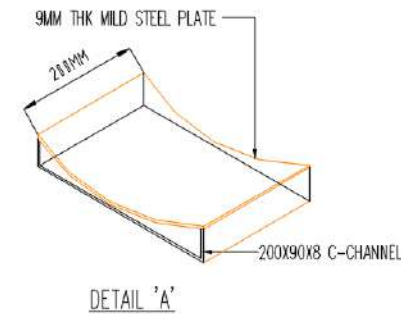
# CHILLED WATER PIPE (ABOVE 350mm):



FOR PIPES  $\phi 350\text{MM}$  TO  $\phi 500$



FOR PIPES  $\phi 550$  TOP 750 MM



DETAIL 'A'



TABLE-A

| PIPE SIZE $\phi$    | TOP CHANNEL     |     | EXPANSION ANCHOR<br>(HILTI HSA) | HANGER ROD<br>D | BOTTOM-CHANNEL  | U-BOLT | MAX<br>SPAN |
|---------------------|-----------------|-----|---------------------------------|-----------------|-----------------|--------|-------------|
| ( $\phi\text{MM}$ ) | K               | Z   | (MM)                            | (MM)            | (MM)            | (MM)   | (MM)        |
| $\phi 350-500$      | 150X75X9TX12.5T | 300 | M12X110X2                       | 15              | 150X75X9TX12.5T | 12     | 5.0         |
| $\phi 501-750$      | 150X75X9TX12.5T | 450 | M12X110X3                       | 20              | 150X75X9TX12.5T | 12     | 5.0         |

# Pipe Weight Calculation :



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Email: namleong@singnet.com.sg Website: www.namleong.com.sg

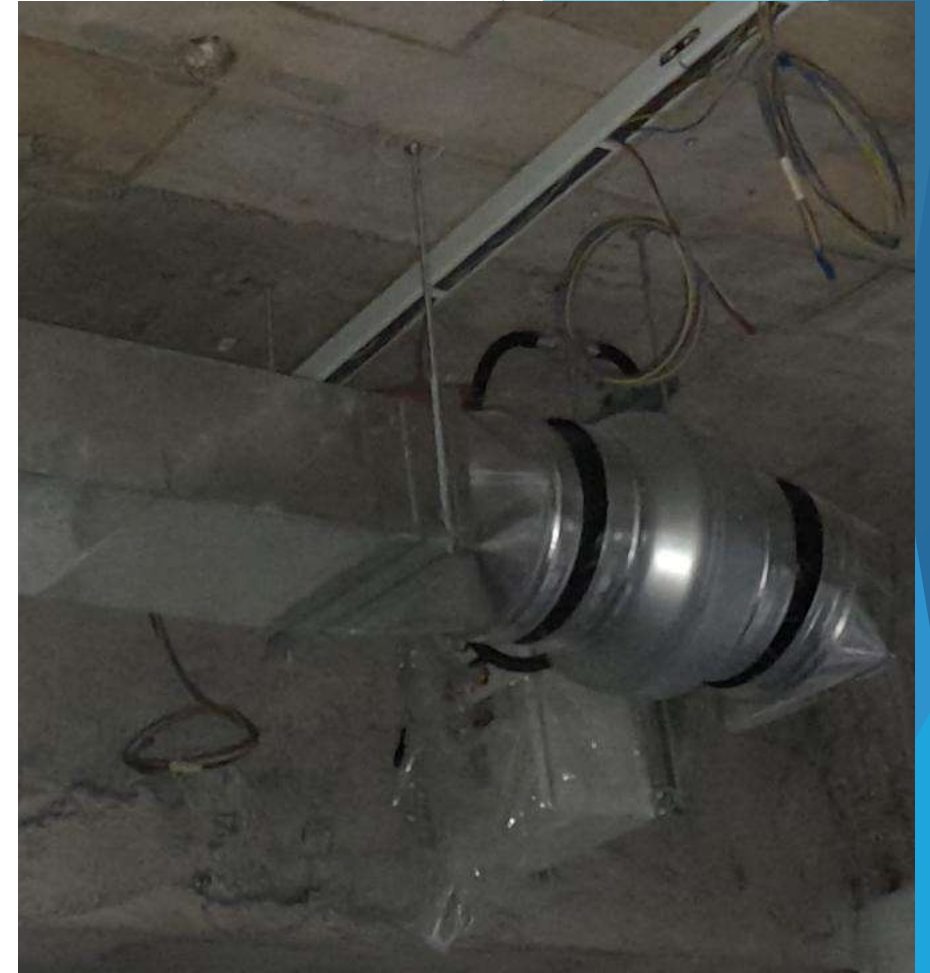
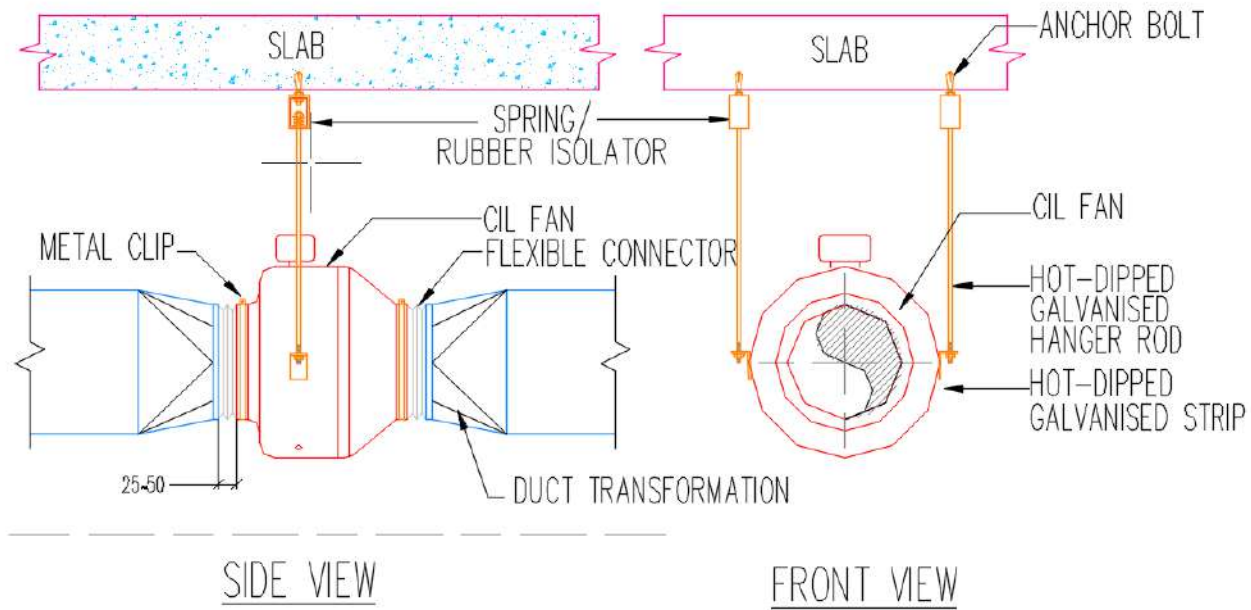


Estimated Pipe Weight

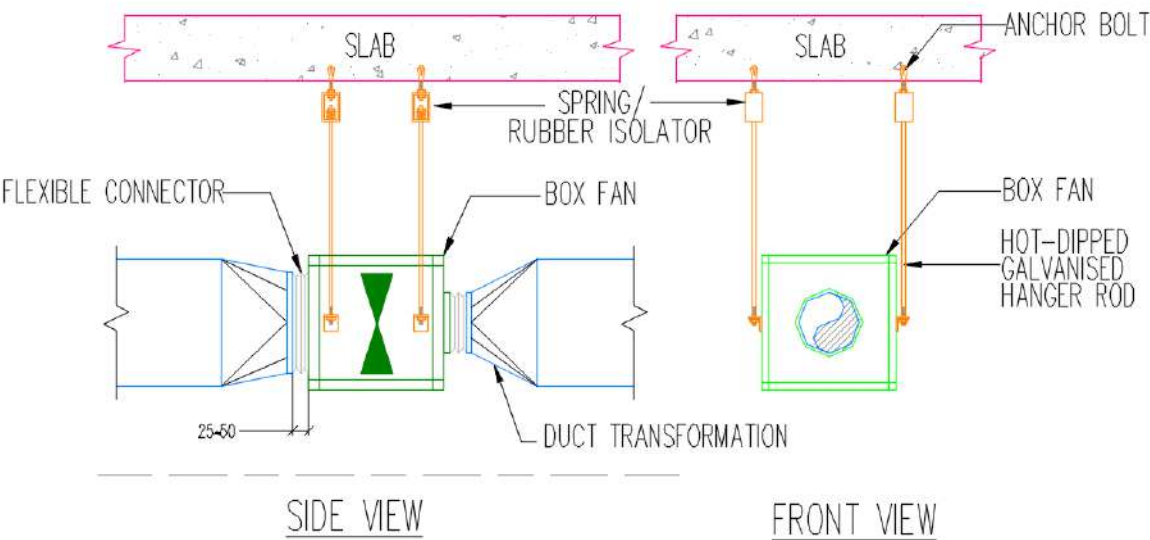
| Nominal<br>Pipe<br>Size<br>(mm) | Carrier Pipe<br>Standard | Approximate<br>Weight of<br>Carrier Pipes<br>(kg/m) | GLENSUL<br>Series No. | Casing<br>Size<br>(mm) | Estimated<br>Weight for<br>Pipe + Water<br>(kg/m) | Estimated<br>Weight for<br>Pipe + insulation<br>(kg/m) | Estimated<br>Weight for<br>Pipe + insulation + water<br>(kg/m) |
|---------------------------------|--------------------------|-----------------------------------------------------|-----------------------|------------------------|---------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------|
| 500                             | JIS G3452                | 594.4                                               | 65                    | 635                    | 1726.4                                            | 701                                                    | 1843                                                           |
| 450                             | JIS G3452                | 525.0                                               | 65                    | 560                    | 1443.0                                            | 634                                                    | 1552                                                           |
| 400                             | JIS G3452                | 465.6                                               | 65                    | 530                    | 1184.6                                            | 561                                                    | 1280                                                           |
| 350                             | JIS G3452                | 406.2                                               | 65                    | 475                    | 950.2                                             | 491                                                    | 1035                                                           |
| 300                             | JIS G3452                | 318.0                                               | 65                    | 450                    | 756.0                                             | 399                                                    | 837                                                            |
| 250                             | JIS G3452                | 254.4                                               | 65                    | 400                    | 559.4                                             | 326                                                    | 631                                                            |
| 200                             | JIS G3452                | 180.6                                               | 65                    | 350                    | 377.8                                             | 243                                                    | 440                                                            |
| 600                             | API 5L B Sch 40          | 1532.6                                              | 65                    | 735                    | 3090.6                                            | 1069                                                   | 3247                                                           |
| 500                             | API 5L B Sch 40          | 1100.6                                              | 65                    | 635                    | 2176.6                                            | 1217                                                   | 2293                                                           |
| 450                             | API 5L B Sch 40          | 934.9                                               | 65                    | 560                    | 1799.9                                            | 1044                                                   | 1900                                                           |
| 400                             | API 5L B Sch 40          | 739.9                                               | 65                    | 530                    | 1423.9                                            | 836                                                    | 1520                                                           |
| 350                             | API 5L B Sch 40          | 567.3                                               | 65                    | 475                    | 1091.3                                            | 652                                                    | 1170                                                           |
| 300                             | API 5L B Sch 40          | 478.3                                               | 65                    | 450                    | 912.3                                             | 558                                                    | 992                                                            |
| 1000                            | API 5L B Sch XS          | 1886.4                                              | 75                    | 1200                   | 6560.4                                            | 2235                                                   | 6010                                                           |
| 950                             | API 5L B Sch Std         | 1347.4                                              | 65                    | 1100                   | 5564.4                                            | 1856                                                   | 5873                                                           |
| 900                             | API 5L B Sch Std         | 1275.4                                              | 65                    | 1050                   | 5050.4                                            | 1579                                                   | 5345                                                           |
| 850                             | API 5L B Sch Std         | 1204.9                                              | 65                    | 1000                   | 4589.9                                            | 1417                                                   | 4782                                                           |
| 800                             | API 5L B Sch Std         | 1133.0                                              | 65                    | 950                    | 4103.0                                            | 1336                                                   | 4306                                                           |
| 750                             | API 5L B Sch Std         | 1061.1                                              | 65                    | 900                    | 3663.1                                            | 1254                                                   | 3856                                                           |
| 700                             | API 5L B Sch Std         | 989.2                                               | 65                    | 850                    | 3242.2                                            | 1173                                                   | 3429                                                           |
| 650                             | API 5L B Sch Std         | 917.3                                               | 65                    | 800                    | 2853.3                                            | 1091                                                   | 3027                                                           |
| 600                             | API 5L B Sch Std         | 846.7                                               | 65                    | 735                    | 2492.7                                            | 1003                                                   | 2640                                                           |
| 550                             | API 5L B Sch Std         | 772.9                                               | 65                    | 635                    | 1829.9                                            | 819                                                    | 1946                                                           |
| 450                             | API 5L B Sch Std         | 631.0                                               | 65                    | 560                    | 1535.0                                            | 740                                                    | 1644                                                           |
| 400                             | API 5L B Sch Std         | 559.6                                               | 65                    | 530                    | 1267.6                                            | 655                                                    | 1353                                                           |
| 350                             | API 5L B Sch Std         | 488.0                                               | 65                    | 475                    | 1021.0                                            | 573                                                    | 1106                                                           |
| 300                             | API 5L B Sch Std         | 443.2                                               | 65                    | 450                    | 891.2                                             | 523                                                    | 961                                                            |
| 250                             | API 5L B Sch Std         | 361.7                                               | 65                    | 400                    | 666.7                                             | 433                                                    | 738                                                            |
| 200                             | API 5L B Sch Std         | 255.3                                               | 65                    | 350                    | 449.3                                             | 317                                                    | 511                                                            |
| 150                             | API 5L B Sch Std         | 169.8                                               | 50                    | 265                    | 291.6                                             | 213                                                    | 325                                                            |
| 125                             | API 5L B Sch Std         | 130.8                                               | 50                    | 240                    | 207.6                                             | 171                                                    | 249                                                            |
| 100                             | API 5L B Sch Std         | 90.5                                                | 50                    | 215                    | 145.5                                             | 126                                                    | 175                                                            |
| 80                              | API 5L B Sch Std         | 67.7                                                | 50                    | 190                    | 96.7                                              | 93                                                     | 122                                                            |
| 65                              | API 5L B Sch Std         | 51.8                                                | 50                    | 170                    | 70.8                                              | 73                                                     | 92                                                             |
| 50                              | API 5L B Sch Std         | 32.6                                                | 40                    | 140                    | 45.8                                              | 50                                                     | 63                                                             |
| 40                              | API 5L B Sch Std         | 24.3                                                | 40                    | 125                    | 32.3                                              | 39                                                     | 47                                                             |
| 32                              | API 5L B Sch Std         | 20.3                                                | 40                    | 115                    | 26.3                                              | 34                                                     | 40                                                             |
| 25                              | API 5L B Sch Std         | 15.0                                                | 40                    | 115                    | 18.0                                              | 29                                                     | 32                                                             |
| 20                              | API 5L B Sch Std         | 10.1                                                | 40                    | 100                    | 12.1                                              | 22                                                     | 24                                                             |
| 15                              | API 5L B Sch Std         | 7.6                                                 | 40                    | 100                    | 8.6                                               | 20                                                     | 21                                                             |
| 150                             | BS1387 Class B           | 118.2                                               | 50                    | 265                    | 232.2                                             | 162                                                    | 276                                                            |
| 125                             | BS1387 Class B           | 60.6                                                | 50                    | 240                    | 178.6                                             | 140                                                    | 210                                                            |
| 100                             | BS1387 Class B           | 73.2                                                | 50                    | 215                    | 126.2                                             | 102                                                    | 164                                                            |
| 80                              | BS1387 Class B           | 50.2                                                | 50                    | 190                    | 81.2                                              | 75                                                     | 100                                                            |
| 65                              | BS1387 Class B           | 38.6                                                | 50                    | 170                    | 60.6                                              | 60                                                     | 82                                                             |
| 50                              | BS1387 Class B           | 30.2                                                | 40                    | 140                    | 43.2                                              | 47                                                     | 60                                                             |
| 40                              | BS1387 Class B           | 21.4                                                | 40                    | 125                    | 30.4                                              | 36                                                     | 45                                                             |
| 32                              | BS1387 Class B           | 18.6                                                | 40                    | 115                    | 24.0                                              | 32                                                     | 38                                                             |
| 25                              | BS1387 Class B           | 14.5                                                | 40                    | 115                    | 18.5                                              | 28                                                     | 32                                                             |
| 20                              | BS1387 Class B           | 9.4                                                 | 40                    | 100                    | 12.4                                              | 21                                                     | 24                                                             |
| 15                              | BS1387 Class B           | 7.3                                                 | 40                    | 100                    | 8.3                                               | 19                                                     | 20                                                             |
| 150                             | BS1387 Class C           | 127.8                                               | 50                    | 265                    | 239.8                                             | 172                                                    | 284                                                            |
| 125                             | BS1387 Class C           | 107.4                                               | 50                    | 240                    | 185.4                                             | 146                                                    | 226                                                            |
| 100                             | BS1387 Class C           | 87.0                                                | 50                    | 215                    | 138.0                                             | 116                                                    | 167                                                            |
| 80                              | BS1387 Class C           | 61.8                                                | 50                    | 190                    | 90.8                                              | 87                                                     | 116                                                            |
| 65                              | BS1387 Class C           | 47.6                                                | 50                    | 170                    | 68.6                                              | 69                                                     | 90                                                             |
| 50                              | BS1387 Class C           | 37.1                                                | 40                    | 140                    | 50.1                                              | 54                                                     | 67                                                             |
| 40                              | BS1387 Class C           | 28.2                                                | 40                    | 125                    | 34.2                                              | 41                                                     | 49                                                             |
| 32                              | BS1387 Class C           | 22.8                                                | 40                    | 115                    | 27.8                                              | 37                                                     | 42                                                             |
| 25                              | BS1387 Class C           | 17.6                                                | 40                    | 115                    | 20.6                                              | 32                                                     | 35                                                             |
| 20                              | BS1387 Class C           | 11.2                                                | 40                    | 100                    | 13.2                                              | 23                                                     | 25                                                             |
| 15                              | BS1387 Class C           | 8.6                                                 | 40                    | 100                    | 9.6                                               | 21                                                     | 22                                                             |

# Equipment Supporting Detail:

## MV FAN(CIL):



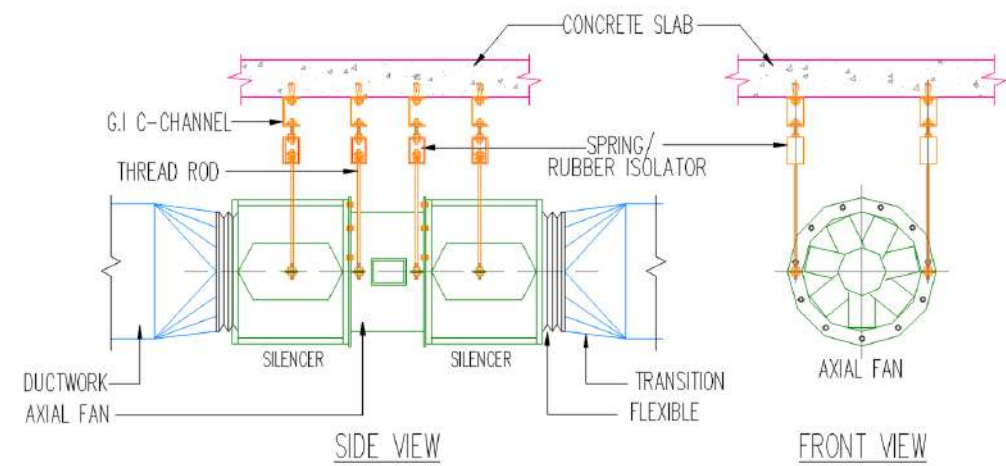
MV FAN(BOX):



ISOLATOR  
ROD  
MV FAN

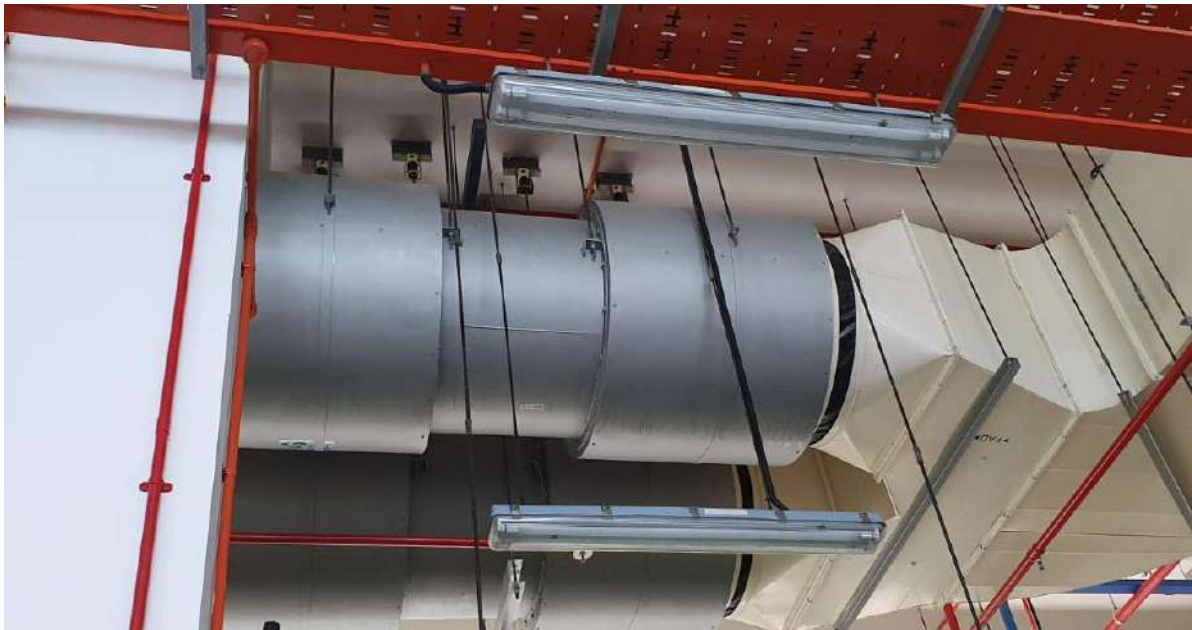


MV FAN (AXIAL):

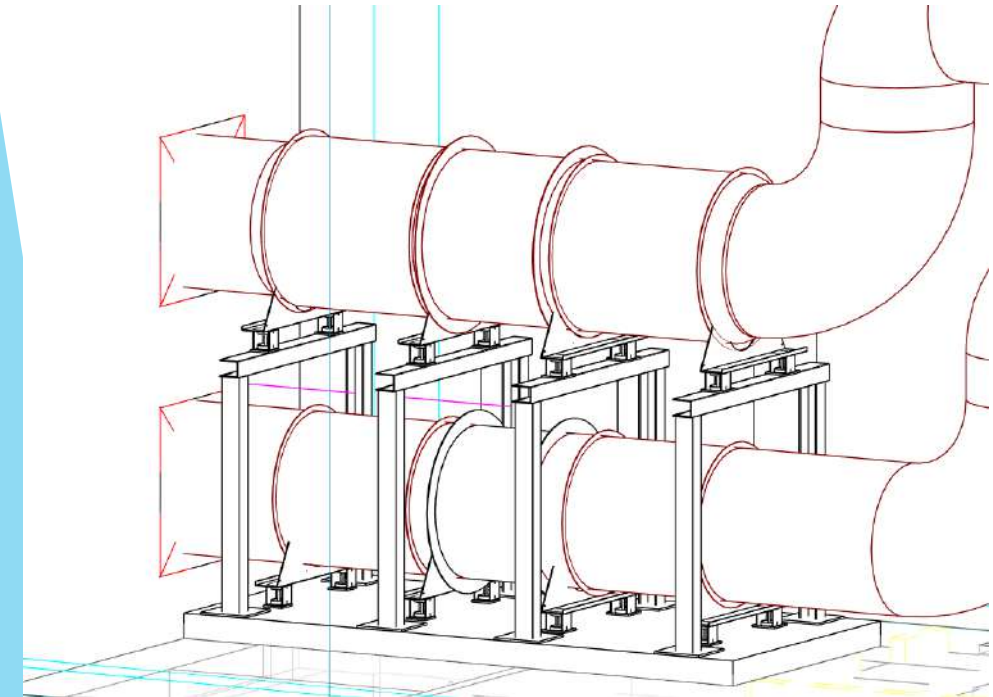
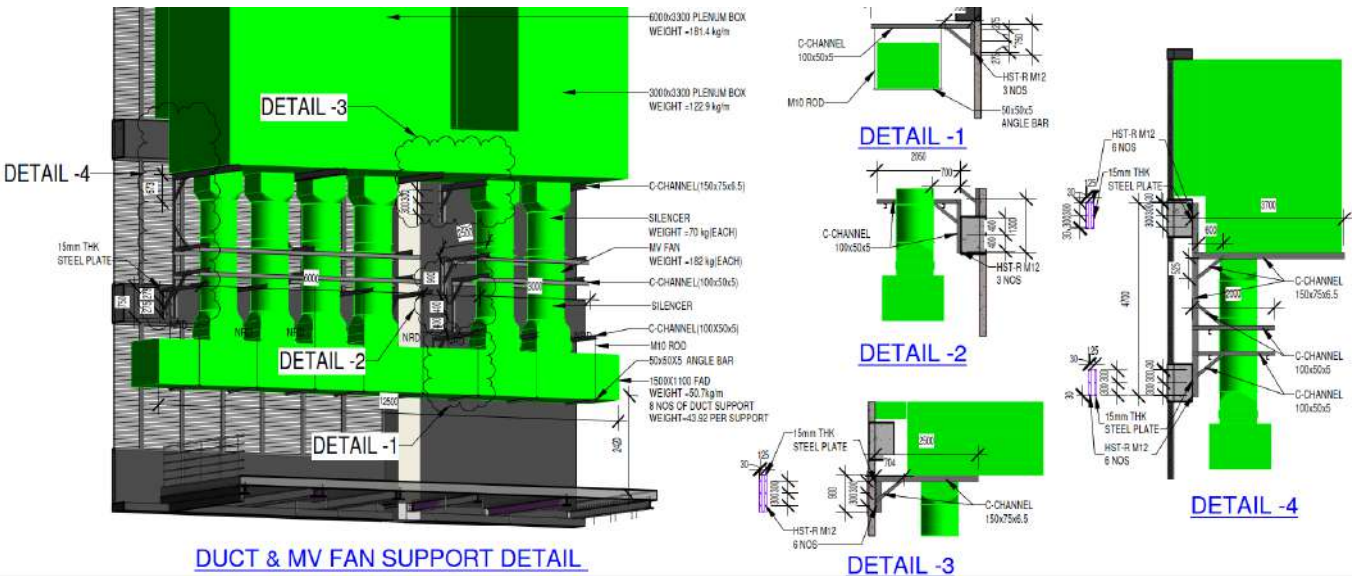


HANGER SIZING

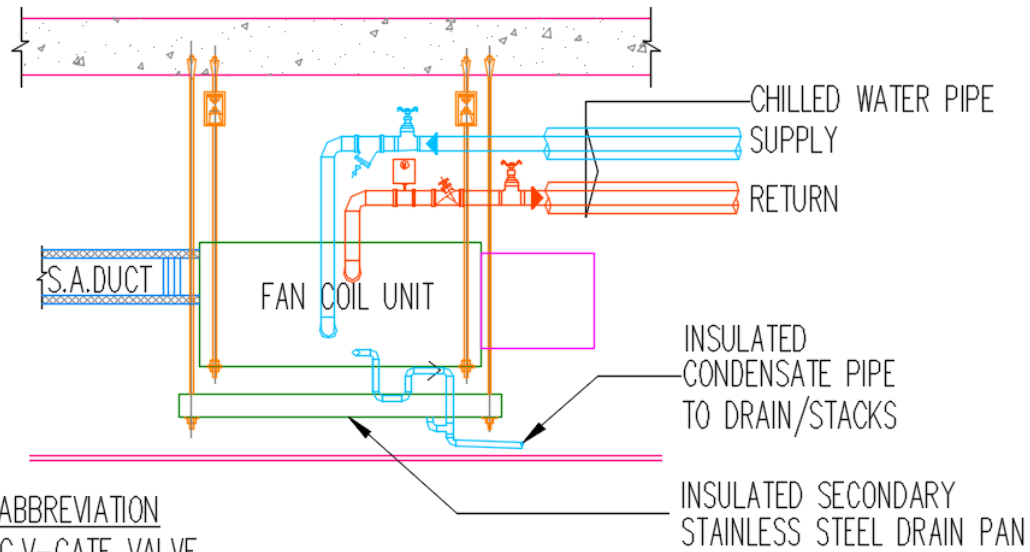
| FAN DIMENSION<br>ϕ mm( inch ) | FANS HANGER AND SUPPORT DIMENSION |                     |                              |
|-------------------------------|-----------------------------------|---------------------|------------------------------|
|                               | HANGER ROD<br>( ϕ mm )            | C-CHANNEL<br>( mm ) | ANCHOR BOLT<br>( mm x Nos. ) |
| 305 (12)                      | 10                                | 100 x 50 x 5        | M10 x 4                      |
| 381 (15)                      | 10                                | 100 x 50 x 5        | M10 x 4                      |
| 481 (19)                      | 10                                | 100 x 50 x 5        | M10 x 4                      |
| 533 (21)                      | 12                                | 100 x 50 x 5        | M12 x 4                      |
| 610 (24)                      | 12                                | 100 x 50 x 5        | M12 x 4                      |
| 762 (30)                      | 16                                | 100 x 50 x 5        | M10 x 2 x 4                  |
| 965 (38)                      | 16                                | 125 x 65 x 6        | M12 x 2 x 4                  |
| 1219 (48)                     | 18                                | 125 x 65 x 6        | M16 x 2 x 4                  |
| 1524 (60)                     | 18                                | 150 x 75 x 8        | M16 x 2 x 4                  |
| 1905 (75)                     | 18                                | 150 x 75 x 8        | M16 x 2 x 4                  |



MV FAN(AXIAL):



## FCU(Ducted):



### ABBREVIATION

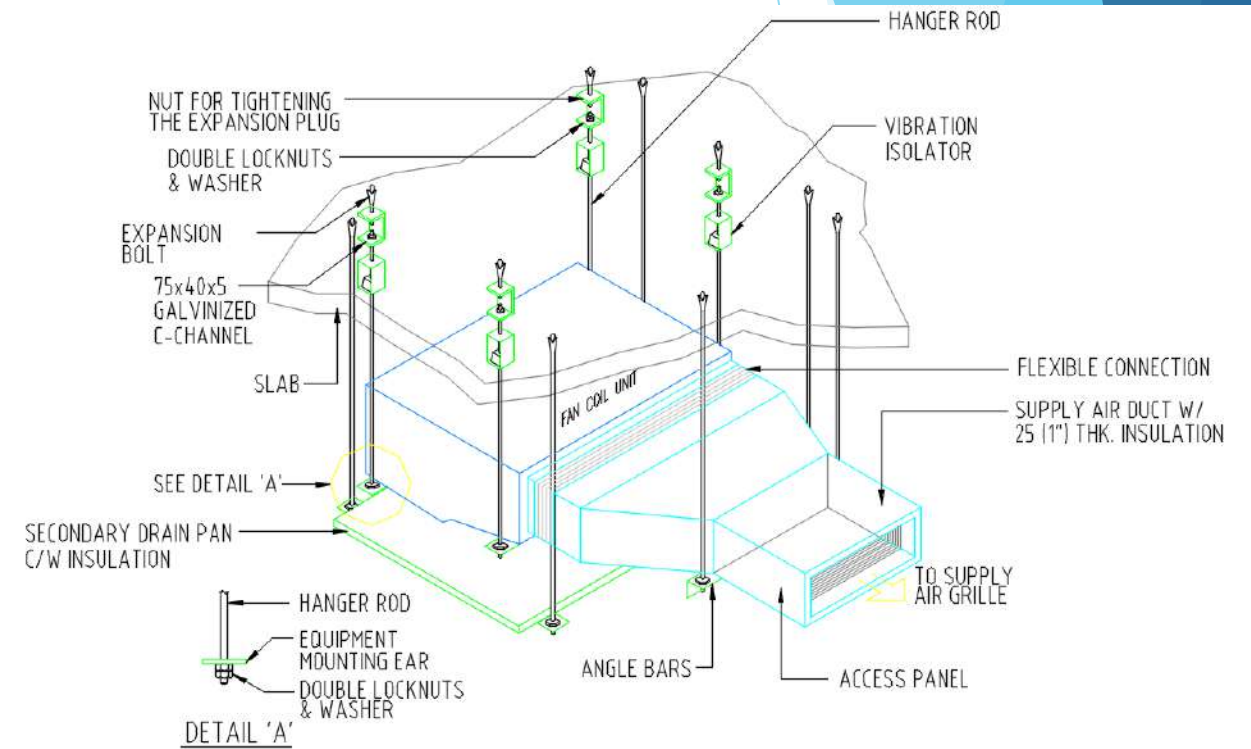
G.V-GATE VALVE

M.V-MODULATING VALVE

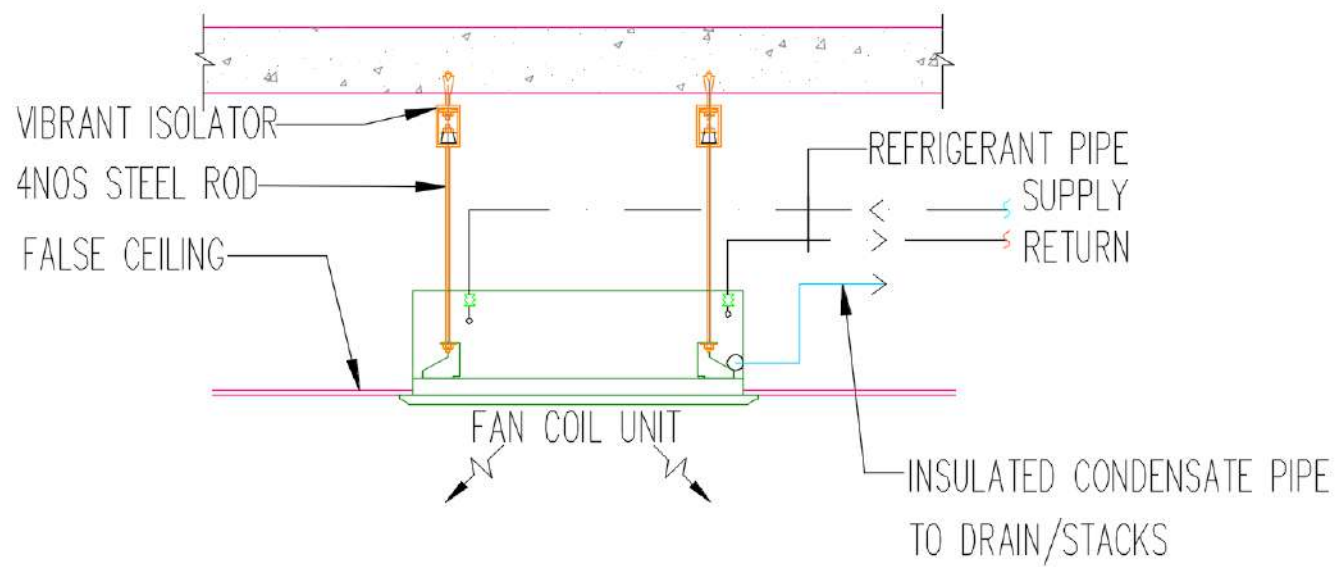
F.C-FLEXIBLE CONNECTOR

S-Y-STRAINER

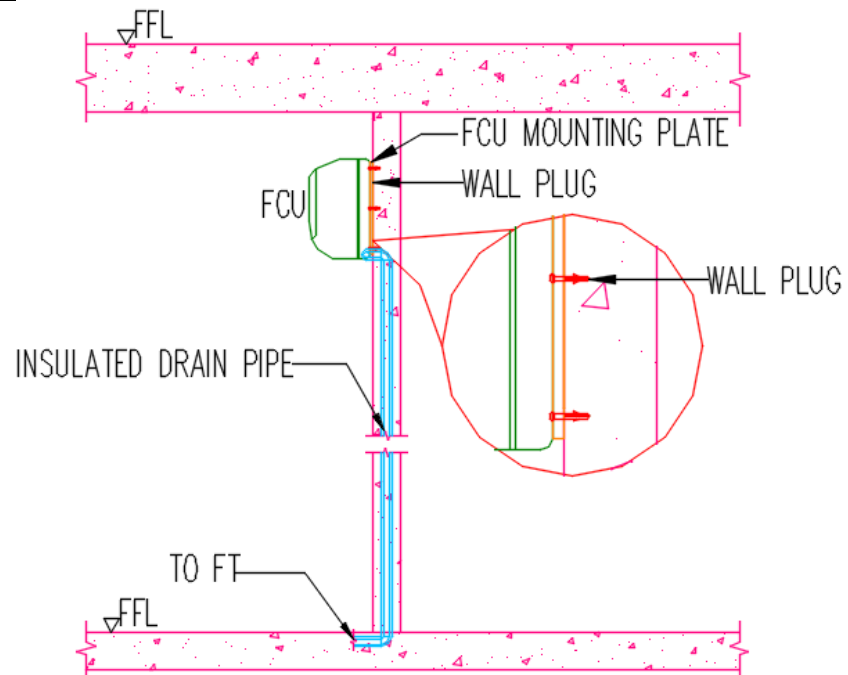
B.V-BALANCING VALVE



## FCU(Cassette):

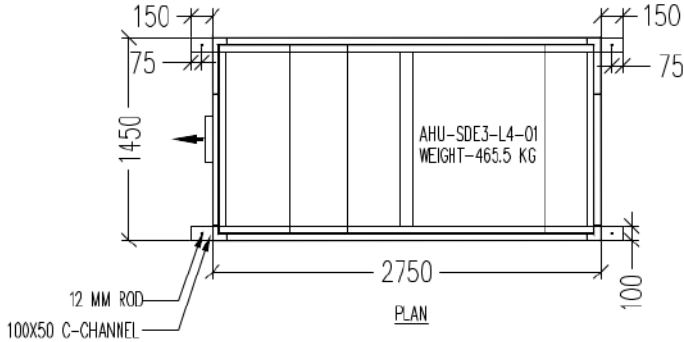
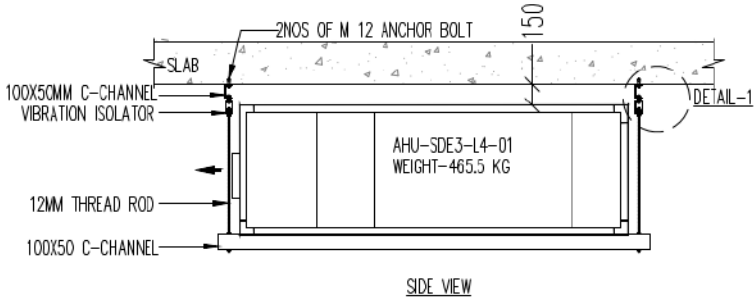


## FCU(Wall mounted):



# Equipment Installation

## AHU:

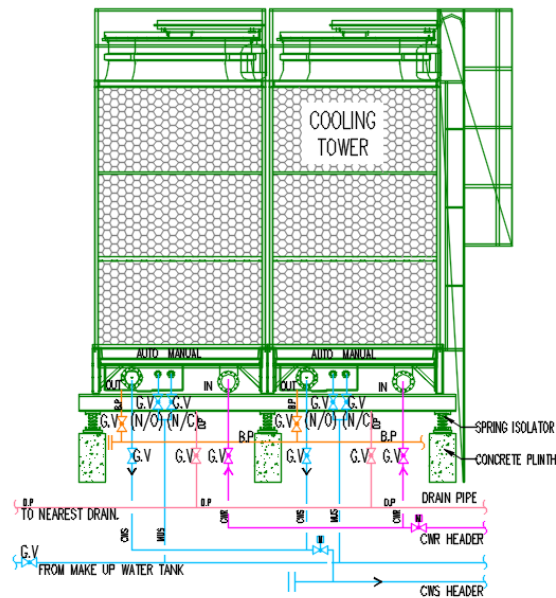


INSTALLATION DETAIL FOR HANGING AHU

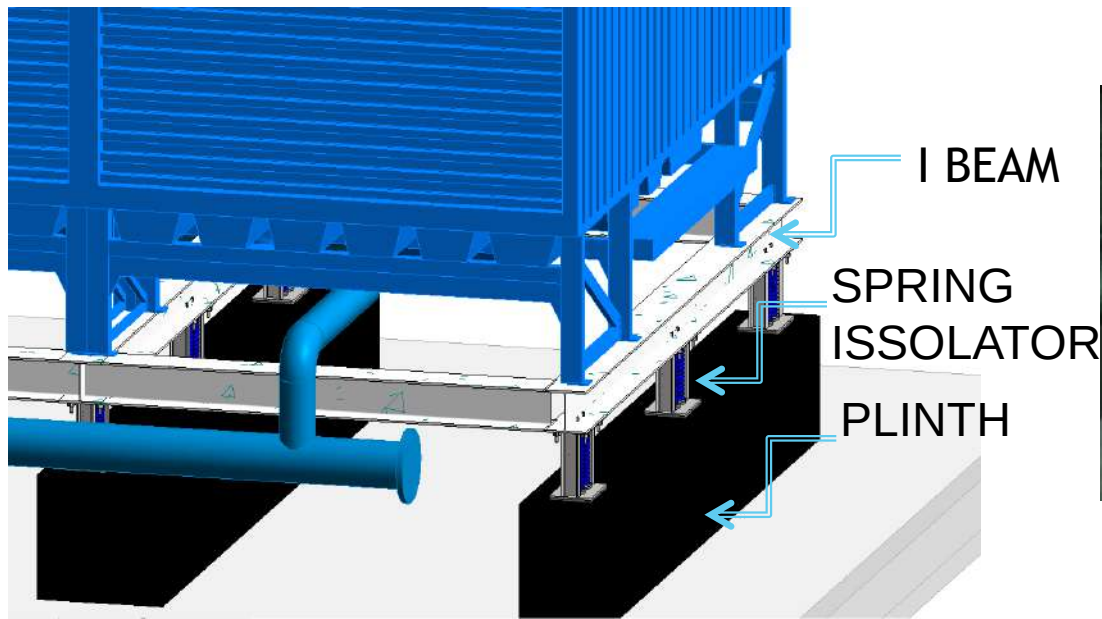


# Equipment Installion

## COOLING TOWER:

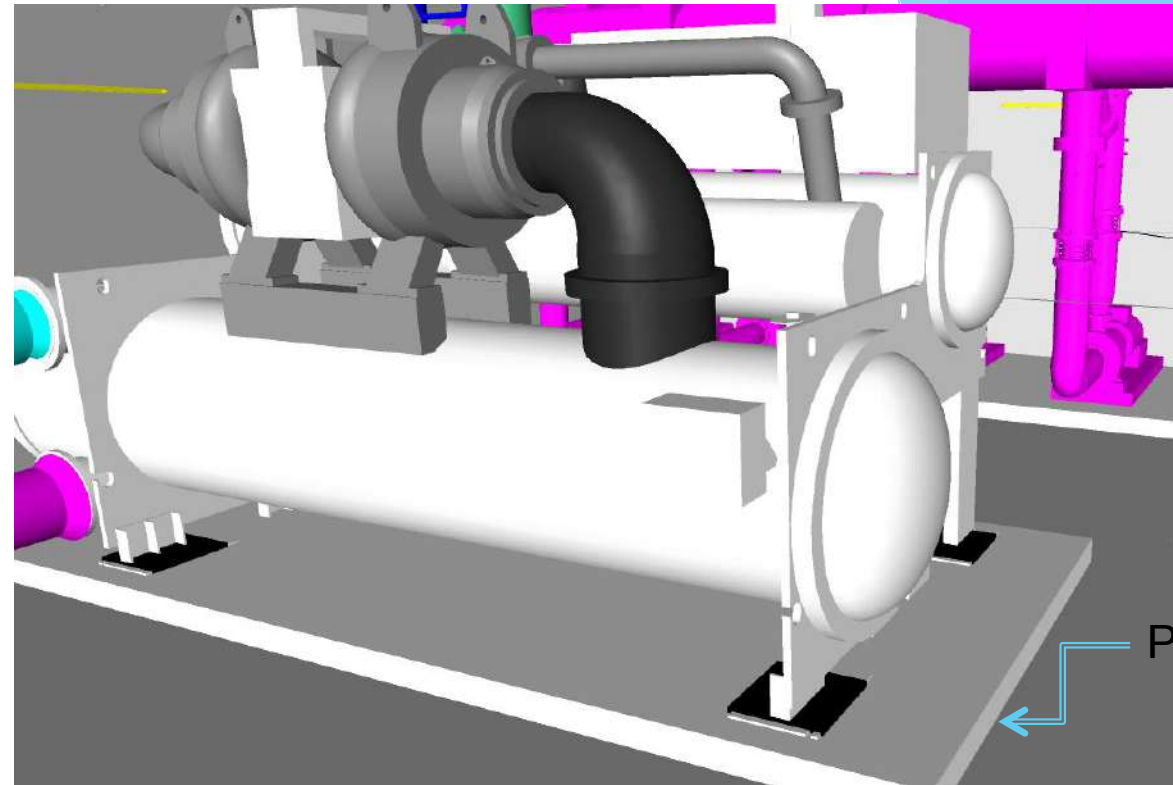
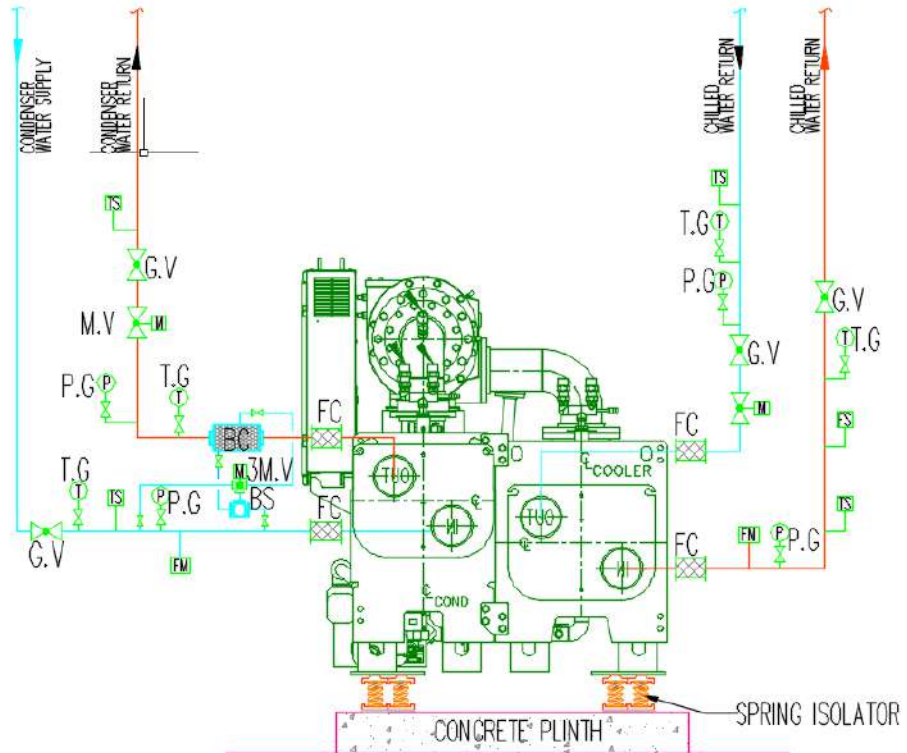


ABBREVIATION  
G.V.-GATE VALVE  
CWS-CONDENSER WATER SUPPLY  
CNR-CONDENSER WATER RETURN  
MUS-MAKE UP WATER SUPPLY  
B.P.-BALANCING PIPE  
D.P.-DRAIN PIPE



# Equipment Installion

## CHILLER:



## INERTIA BLOCK :

